Chapter 4

Approximating hedging strategies by neural networks

The previous chapter established that, under the hypotheses (A1) to (A6), the hedging functional π admits an optimal strategy on a general probability space. This chapter turns to the second question: can the optimal hedge be approximated on a general Ω by neural network strategies? On a finite probability space, Bühler et al. [1, Proposition 4.3] prove that the network values $\pi_M(X)$ decrease to $\pi(X)$ as the network capacity M grows. We identify where that argument uses finiteness of Ω, show by counterexample that the result fails in general. We then restore $\pi_M \to \pi$ under (A1) and (A6), the essential hypotheses of Chapter 3 to prove attainment. Combined with the attainment theorem of Chapter 3, this yields ε-optimal network strategies for an attained optimum.

4.1 Neural networks, market setup and the approximate problem

The market and the hedging functional

We continue in the market setting of Chapter 2 (information process I, mid-price S, unconstrained strategies $\mathcal{H}^{\mathrm{u}}$, constraints via $H \circ \delta$, costs C_T and gains W) and write

$$\pi(X) := \inf_{\delta \in \mathcal{H}^{\mathrm{u}}} \rho\big(X + W(H \circ \delta)\big) \tag{1}$$

for the hedging functional of Chapter 3. Until the repair we do not impose (A1) to (A6); those hypotheses enter only when we restore $\pi_M \to \pi$ on general Ω.

Neural networks

A feed forward neural network with activation function $\sigma\colon \mathbb{R}\to\mathbb{R}$ is a map $F = W_L \circ F_{L-1} \circ \cdots \circ F_1\colon \mathbb{R}^{d_0} \to \mathbb{R}^{d_1}$ with $F_j = \sigma \circ W_j$ for affine maps W_j, where σ acts componentwise; see Bühler et al. [1, Definition 2]. We denote by $\mathcal{NN}_{\infty,d_0,d_1}$ the set of all such networks and fix, as in [1, Section 4.1], an increasing exhaustion $\mathcal{NN}_{M,d_0,d_1} \subseteq \mathcal{NN}_{M+1,d_0,d_1}$ with $\bigcup_{M\in\mathbb{N}} \mathcal{NN}_{M,d_0,d_1} = \mathcal{NN}_{\infty,d_0,d_1}$, each $\mathcal{NN}_{M,d_0,d_1}$ parametrized by finitely many weights and containing the zero network, and such that networks with fewer inputs embed by zero weight connections. Throughout, the activation function σ is measurable, bounded and nonconstant. The analytical basis of the approximation theory is the following universal approximation theorem of Hornik [2, Theorem 1].

Theorem 4.1. *Let $\sigma\colon \mathbb{R}\to\mathbb{R}$ be measurable, bounded and nonconstant, and let $d_0 \in \mathbb{N}$. Then for every finite measure μ on $\mathbb{R}^{d_0}$ and every $p \in [1,\infty)$, the set $\mathcal{NN}_{\infty,d_0,1}$ is dense in $L^p(\mathbb{R}^{d_0},\mu)$. In particular, applying the result componentwise, $\mathcal{NN}_{\infty,d_0,d}$ is dense in $L^p(\mathbb{R}^{d_0},\mu;\mathbb{R}^d)$ for every $d \in \mathbb{N}$.*

The deep hedging approach replaces the infinite-dimensional strategy class $\mathcal{H}^{\mathrm{u}}$ by finite-dimensional subclasses of network strategies. For each capacity $M \in \mathbb{N}$, following Bühler et al. [1, proof of Proposition 4.3], we set

$$\mathcal{H}_M := \big\{\delta \in \mathcal{H}^{\mathrm{u}} : \delta_k = F_k(I_0,\dots,I_k) \text{ with } F_k \in \mathcal{NN}_{M,r(k+1),d}\big\}.$$

Thus a strategy belongs to $\mathcal{H}_M$ if and only if each hedge ratio δ_k is given by a network of complexity at most M applied to the information $(I_0,\dots,I_k)$ available at time t_k. We do not include the previous position δ_{k-1} as an additional network input. Indeed, each δ_k is $\mathcal{F}_k$-measurable with $\mathcal{F}_k = \sigma(I_0,\dots,I_k)$, and δ_{k-1} is a function of $(I_0,\dots,I_{k-1},\delta_{k-2})$; recursively, every δ_k is already a function of $(I_0,\dots,I_k)$ alone, so feeding δ_{k-1} back would not enlarge the class of adapted strategies. The associated approximate hedging problem is

$$\pi_M(X) := \inf_{\delta\in\mathcal{H}_M} \rho\big(X + W(H\circ\delta)\big). \tag{2}$$

The question of this chapter is whether $\pi_M(X)$ approaches $\pi(X)$ as $M\to\infty$. The following elementary observation reduces that question to a one-sided estimate.

Lemma 4.2. *For every X and every $M \in \mathbb{N}$, one has*

$$\pi_M(X) \geq \pi_{M+1}(X) \geq \pi(X).$$

In particular, $\lim_{M\to\infty} \pi_M(X)$ exists in $[-\infty,\infty]$ and dominates $\pi(X)$.

Proof. By definition, strategies in $\mathcal{H}_M$ are processes $\delta = (\delta_k)_{k=0,\dots,n-1}$ with

$$\delta_k = F_k(I_0,\dots,I_k), \qquad k = 0,\dots,n-1,$$

for network maps F_k. Therefore, each δ_k is $\mathcal{F}_k$-measurable and δ is adapted. With the nested exhaustion of the network classes one has $\mathcal{H}_M \subseteq \mathcal{H}_{M+1} \subseteq \mathcal{H}^{\mathrm{u}}$, so the infima defining π_M, π_{M+1} and π in (1) and (2) are taken over increasing sets, which gives both inequalities. The limit exists because the sequence $(\pi_M(X))_{M\in\mathbb{N}}$ is nonincreasing and bounded below by $\pi(X)$ in $[-\infty, \infty]$. □

It remains to decide whether the limit equals $\pi(X)$.

4.2 The finite-space theorem and what breaks

Bühler et al., Proposition 4.3, prove the following on a finite probability space. In the proof we flag each use of finiteness of Ω as (F1) to (F3); these are exactly the points that need attention on a general probability space.

Proposition 4.3. *Suppose the probability space $\Omega = \{\omega_1, \dots, \omega_N\}$ is finite with $\mathbb{P}[\{\omega_i\}] > 0$ for all i, and ρ is a convex risk measure. Then for every claim X one has $\lim_{M\to\infty} \pi_M(X) = \pi(X)$.*

Proof. By monotonicity of $(\pi_M)_{M\in\mathbb{N}}$ (Lemma 4.2) it suffices to show that for every $\varepsilon > 0$ some M satisfies $\pi_M(X) \leq \pi(X) + \varepsilon$.

Choose $\delta \in \mathcal{H}^{\mathrm{u}}$ with

$$\rho\big(X + W(H \circ \delta)\big) \leq \pi(X) + \frac{\varepsilon}{2}. \tag{3}$$

Write $\delta_k = f_k(I_0, \dots, I_k)$ for Borel maps $f_k : \mathbb{R}^{r(k+1)} \to \mathbb{R}^d$.

(F1) Integrability of the hedge ratios. Because Ω is finite, each δ_k takes finitely many values and every component of f_k lies in $L^1(\mu_k)$ where μ_k is the law of $(I_0, \dots, I_k)$. Finally, Theorem 4.1 yields networks $F_{k,m} \to f_k$ in $L^1(\mu_k)$.

(F2) From L^1 to pointwise convergence. On a finite Ω, L^1 convergence implies pointwise convergence: $\delta_k^m := F_{k,m}(I_0, \dots, I_k) \to \delta_k$ for every $\omega \in \Omega$.

Set $G_m := ((H \circ \delta^m) \cdot S)_T$ and $C_m := C_T(H \circ \delta^m)$. Since H and the stochastic integral are continuous, $G_m \to G$ pointwise. Furthermore, using upper semicontinuity of the c_k and continuity of H again, one has $\limsup_m C_m = \overline{C} \leq C := C_T(H \circ \delta)$ pointwise.

(F3) Continuity of the risk measure. On the finite space Ω, every random variable may be identified with a vector in $\mathbb{R}^N$, so ρ may be viewed as a convex map $\mathbb{R}^N \to \mathbb{R}$ and is therefore continuous. Continuity and monotonicity of ρ give

$$\lim_{m\to\infty} \rho(X + G_m - C_m) = \rho(X + G - \overline{C}) \leq \rho(X + G - C) \leq \pi(X) + \frac{\varepsilon}{2}$$

by (3). Pick m large enough so that the left-hand side is $\leq \pi(X)+\varepsilon$. Since the network families are nested with $\bigcup_{M\in\mathbb{N}} \mathcal{NN}_{M,r(k+1),d} = \mathcal{NN}_{\infty,r(k+1),d}$, there exists M_m such that $\delta^m \in \mathcal{H}_M$ for all $M \geq M_m$, and therefore $\pi_M(X) \leq \pi(X) + \varepsilon$ for all such M. □

What breaks on a general probability space

Remark 4.4. (F1): hedge ratios need not be integrable. On general Ω, the representing functions f_k need not lie in $L^1(\mu_k)$, and then Theorem 4.1 cannot even be invoked: f_k is not an element of the space in which networks are dense.

Example 4.5. Let I_0 have a standard Cauchy law μ_0 and $\mathcal{F}_0 = \sigma(I_0)$. The strategy $\delta_0 = f(I_0)$ with $f(x) = |x|$ is perfectly admissible as a measurable adapted process, but $f \notin L^1(\mu_0)$, so no sequence of μ_0-integrable functions (in particular no sequence of networks with bounded activation, which are bounded functions) converges to f in $L^1(\mu_0)$.

Remark 4.6. (F2): only subsequential almost sure convergence (harmless). On general Ω, $L^1(\mathbb{P})$-convergence yields almost sure convergence only along a subsequence. This is harmless for the argument: the proof only requires *one* good approximating strategy, so we may freely pass to subsequences.

Remark 4.7. (F3): the risk measure is not continuous along almost sure convergence. This is the decisive failure. On $\mathbb{R}^N$, convexity of ρ implies continuity. On an infinite-dimensional domain, a convex risk measure need not be continuous along almost sure convergence. The direction needed in the final step of the proof is an *upper* semicontinuity:

$$\limsup_{m\to\infty} \rho(Y_m) \le \rho(Y) \qquad \text{whenever } Y_m \to Y \ \mathbb{P}\text{-a.s.}$$

This fails even for the entropic risk measure, which satisfies the half-bounded *lower* semicontinuity (A3) of Chapter 3.

Example 4.8. Work on $\Omega = (0, 1)$ with the Borel σ-field and Lebesgue measure, let $\lambda > 0$ and let $\rho_\lambda(Y) = \frac{1}{\lambda} \log \mathbb{E}[e^{-\lambda Y}]$ be the entropic risk measure. Set $A_m := (0, e^{-m})$ and

$$Y_m := -\frac{2m}{\lambda}\mathbf{1}_{A_m}.$$

For every $\omega \in (0, 1)$ one has $\omega \notin A_m$ for all $m > \log(1/\omega)$, so $Y_m \to 0$ everywhere. But

$$\mathbb{E}\big[e^{-\lambda Y_m}\big] = 1 - e^{-m} + e^{-m}e^{2m} = 1 - e^{-m} + e^{m} \longrightarrow \infty,$$

so $\rho_\lambda(Y_m) \to \infty$ while $\rho_\lambda(0) = 0$. Small events carrying large losses make the risk explode along an almost surely vanishing sequence.

Failure on a general probability space

We now show that Proposition 4.3 does not extend to a general probability space. The example is a one-period market with a perfect hedge whose hedge ratio is integrable but has no exponential moments.

Proposition 4.9. *There exist a one-period market on a general probability space, without transaction costs and without trading constraints, an entropic risk measure ρ_λ with arbitrary risk aversion $\lambda > 0$ and a position $X = -Z$ with $Z \in L^1$ such that*

$$\pi(X) \leq 0 \qquad \text{and} \qquad \pi_M(X) = +\infty \quad \text{for every } M \in \mathbb{N}.$$

Moreover, $\rho_\lambda(X + \delta_0(S_1 - S_0)) = +\infty$ for every *strategy of the form $\delta_0 = F(I_0)$ with $F : \mathbb{R} \to \mathbb{R}$ bounded on $(0,1)$; this covers all networks with bounded activation and all networks with continuous activation.*

Proof. Let $\Omega := (0,1) \times \{-1,+1\}$ with $\mathbb{P} := \mathrm{Leb} \otimes (\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_{+1})$, and write $U(\omega) := \omega_1$ and $\xi(\omega) := \omega_2$, so that U is uniform on $(0,1)$, $\mathbb{P}[\xi = \pm 1] = \frac{1}{2}$ and U, ξ are independent. Take one trading period ($n = 1$), information $I_0 := U$, $I_1 := \xi$, hence $\mathcal{F}_0 = \sigma(U)$ and $\mathcal{F}_1 = \sigma(U, \xi)$, one asset ($d = 1$) with

$$S_0 := 1, \qquad S_1 := 1 + \frac{\xi}{2},$$

no costs ($c_k \equiv 0$) and no constraints ($H_0(\delta_0) := \delta_0$). Note that S is a $\mathbb{P}$-martingale and $S_1 > 0$. Define

$$h(u) := \frac{2}{\lambda} \log \frac{1}{u}, \qquad u \in (0,1), \qquad Z := h(U)\frac{\xi}{2}, \qquad X := -Z.$$

Then $\mathbb{E}[|Z|] = \frac{1}{2}\mathbb{E}[h(U)] = \frac{1}{\lambda} < \infty$, so $Z \in L^1$.

The perfect hedge. The strategy $\delta_0 := h(U)$ is $\mathcal{F}_0$-measurable, and

$$X + \delta_0(S_1 - S_0) = -h(U)\frac{\xi}{2} + h(U)\frac{\xi}{2} = 0,$$

so $\pi(X) \leq \rho_\lambda(0) = 0$. Note that $\delta_0 = h(U) \in L^1$; the optimal strategy is integrable, so Theorem 4.1 applies to it and networks approximate it in $L^1(\mathbb{P})$ and, along a subsequence, almost surely.

Every locally bounded strategy has infinite risk. Let $\delta_0 = F(U)$ with $m := \sup_{u \in (0,1)} |F(u)| < \infty$. Then

$$\rho_\lambda\big(X + F(U)(S_1 - S_0)\big) = \frac{1}{\lambda} \log \mathbb{E}\Big[\exp\Big(\frac{\lambda}{2}\big(h(U) - F(U)\big)\xi\Big)\Big].$$

The integrand is nonnegative, so the expectation is at least as large as its restriction to the event $\{\xi = +1\} \cap \{h(U) > m\}$. On that event one has $h(U) - F(U) \geq h(U) - m > 0$, hence

$$\begin{aligned} \exp\Big(\frac{\lambda}{2}\big(h(U) - F(U)\big)\xi\Big) &= \exp\Big(\frac{\lambda}{2}\big(h(U) - F(U)\big)\Big) \\ &\geq \exp\Big(\frac{\lambda}{2}\big(h(U) - m\big)\Big) = \frac{e^{-\lambda m/2}}{U}. \end{aligned}$$

Independence of U and ξ and $\mathbb{P}[\xi = +1] = \frac{1}{2}$ therefore yield, with $u_m := e^{-\lambda m/2}$,

$$\mathbb{E}\Big[\exp\Big(\frac{\lambda}{2}\big(h(U) - F(U)\big)\xi\Big)\Big] \geq \frac{e^{-\lambda m/2}}{2} \int_0^{u_m} \frac{1}{u}\, du = \infty,$$

so the entropic risk is $+\infty$.

Networks are locally bounded. If the activation σ is bounded, every network $F = W_L \circ F_{L-1} \circ \cdots \circ F_1$ is bounded on all of $\mathbb{R}$, because the output of the last hidden layer lies in a bounded set and W_L is affine. If σ is continuous, F is continuous on $\mathbb{R}$ and therefore bounded on the compact set $[0,1] \supseteq (0,1)$. In either case, every network strategy $\delta_0 = F(I_0)$ has $\sup_{(0,1)} |F| < \infty$, so $\rho_\lambda(X + W(\delta)) = +\infty$ and $\pi_M(X) = +\infty$ for every M. $\square$

4.3 Repair under the Chapter 3 hypotheses

The finite-space approximation fails on general Ω at (F1) and (F3). We restore it under two hypotheses from Chapter 3: (A1), that the liability is essentially bounded ($Z \in L^\infty$), and (A6), that every admissible gain is nonnegative ($W(\delta) \geq 0$ a.s. for all $\delta \in \mathcal{H}$). From now on we restrict attention to optimal certainty equivalent (OCE) risk measures.

The first lemma repairs (F3); the failure of the risk measure to be upper semicontinuous along almost surely convergent sequences. Recall from Chapter 2 that the OCE risk measure associated with a convex nondecreasing loss function $\ell : \mathbb{R} \to \mathbb{R}$ is

$$\rho_\ell(Y) := \inf_{w \in \mathbb{R}} \big\{ w + \mathbb{E}[\ell(-Y - w)] \big\}.$$

Lemma 4.10. *Let $\ell : \mathbb{R} \to \mathbb{R}$ be convex and nondecreasing and let ρ_ℓ be the associated OCE. Let $(Y_m)_{m \in \mathbb{N}}$ and Y be random variables with*

$$Y_m \geq -B \ \mathbb{P}\text{-a.s. for all } m,$$

for some deterministic $B \geq 0$. If $\liminf_{m\to\infty} Y_m \geq Y$ $\mathbb{P}$-a.s., then

$$\limsup_{m \to \infty} \rho_\ell(Y_m) \leq \rho_\ell(Y).$$

Proof. Fix $w \in \mathbb{R}$. Since $-(Y_m + w) \leq B + |w|$ and ℓ is nondecreasing,

$$g_w := \ell(B + |w|) \geq \ell\big(-(Y_m + w)\big),$$

so g_w is a deterministic (hence integrable) constant dominating $\ell\big(-(Y_m+w)\big)$. Extend ℓ to $[-\infty, \infty)$ by $\ell(-\infty) := \lim_{t \to -\infty} \ell(t)$; the extension is continuous and nondecreasing. For any real sequence $(a_m)_{m \in \mathbb{N}}$ with $\sup_m a_m < \infty$ one has $\limsup_m \ell(a_m) \leq \ell(\limsup_m a_m)$. Applying this with $a_m = -(Y_m + w)$

and using $\limsup_m\big(-(Y_m+w)\big) = -\liminf_m(Y_m+w) \le -(Y+w)$ a.s. yields

$$\limsup_{m\to\infty} \ell\big(-(Y_m+w)\big) \le \ell\big(-(Y+w)\big)\ \mathbb{P}\text{-a.s.}.$$

Reverse Fatou applied to the sequence

$$g_w \ge \ell\big(-(Y_m+w)\big)$$

gives

$$\limsup_{m\to\infty} \mathbb{E}\big[\ell\big(-(Y_m+w)\big)\big] \le \mathbb{E}\big[\ell\big(-(Y+w)\big)\big].$$

Since $\rho_\ell(Y_m) \le w + \mathbb{E}[\ell(-(Y_m+w))]$, one obtains $\limsup_m \rho_\ell(Y_m) \le w + \mathbb{E}[\ell(-(Y+w))]$ for every w, and taking the infimum over w completes the proof. $\square$

The failure point (F1) is the integrability of the hedge ratios: on a general probability space, near-optimal strategies need not have integrable hedge ratios, so Theorem 4.1 cannot be applied directly (Example 4.5). Under (A1) and (A6), every admissible objective satisfies $X + W(\eta) = -Z + W(\eta) \ge -\|Z\|_\infty$, which is the deterministic lower bound needed for Lemma 4.10. The next lemma exploits this to show that bounded strategies are value-dense: every strategy can be replaced by a bounded one without worsening the OCE objective by more than an arbitrary $\varepsilon > 0$. In particular, π equals the infimum of the objective over bounded strategies alone.

Lemma 4.11. *Assume (A1) and (A6), and let $X = -Z$. Then for every $\delta \in \mathcal{H}^{\mathrm{u}}$ and every $\varepsilon > 0$ there exists a bounded $\delta' \in \mathcal{H}^{\mathrm{u}}$ with*

$$\rho_\ell\big(X + W(H\circ\delta')\big) \le \rho_\ell\big(X + W(H\circ\delta)\big) + \varepsilon.$$

In particular,

$$\pi(X) = \inf\big\{\rho_\ell\big(X + W(H\circ\delta)\big) : \delta \in \mathcal{H}^{\mathrm{u}},\ \delta \textit{ bounded}\big\}.$$

Proof. Fix $\delta \in \mathcal{H}^{\mathrm{u}}$ and write $\eta := H\circ\delta$, $Y := X + W(\eta)$. Define the componentwise truncations $\delta^{(j)} = (\delta_k^{(j)})_{k=0,\dots,n-1}$ for $j \in \mathbb{N}$ by

$$\delta_k^{(j)} := \begin{cases} \delta_k & \text{if } \delta_k \in [-j,j], \\ -j & \text{if } \delta_k < -j, \\ j & \text{if } \delta_k > j, \end{cases} \qquad k = 0,\dots,n-1$$

(applied componentwise if $d > 1$). Then each $\delta^{(j)}$ is adapted (hence in $\mathcal{H}^{\mathrm{u}}$) and bounded by j. Set $\eta^{(j)} := H\circ\delta^{(j)}$ and $Y_j := X + W(\eta^{(j)})$.

Since there are finitely many time steps, there exists a (random) index $j_0 < \infty$, namely

$$j_0 := \max_{\substack{0\le k\le n-1\\ 1\le i\le d}} \big\lceil |\delta_k^i| \big\rceil,$$

such that $\delta_k^{(j)} = \delta_k$ for all k and all $j \geq j_0$. In particular $\delta_k^{(j)} \to \delta_k$ pointwise for every k. The recursive definition of $H \circ$ then yields $\eta^{(j)} \to \eta$ pointwise, and therefore $Y_j \to Y$ pointwise.

By (A1) and (A6),

$$Y_j = -Z + W(\eta^{(j)}) \geq -\|Z\|_\infty \ \mathbb{P}\text{-a.s.}$$

for every j. Lemma 4.10 with $B = \|Z\|_\infty$ gives

$$\limsup_{j\to\infty} \rho_\ell(Y_j) \leq \rho_\ell(Y).$$

Choose $j_\varepsilon \in \mathbb{N}$ large enough that $\rho_\ell(Y_{j_\varepsilon}) \leq \rho_\ell(Y) + \varepsilon$, and set $\delta' := \delta^{(j_\varepsilon)}$. □

The following theorem combines the two repairs with the monotone sandwich of Lemma 4.2. It is the main result of this chapter.

Theorem 4.12. *Assume the standing market and network setting of Bühler et al., let $\rho = \rho_\ell$ be an OCE risk measure, and assume (A1) and (A6). Then*

$$\lim_{M\to\infty} \pi_M(-Z) = \pi(-Z)$$

in $[-\infty, \infty]$, the convergence being monotone from above.

Proof. Write $X := -Z$. We distinguish the three cases $\pi(X) = +\infty$, $\pi(X) \in \mathbb{R}$ and $\pi(X) = -\infty$.

Case $\pi(X) = +\infty$. By Lemma 4.2, $\pi_M(X) \geq \pi(X) = +\infty$ for every M, so $\pi_M(X) = +\infty$ for all M and the claim is trivial.

Case $\pi(X) \in \mathbb{R}$. By Lemma 4.2 it is enough to show that for every $\varepsilon > 0$ there exists M with $\pi_M(X) \leq \pi(X) + \varepsilon$.

Step 1: a bounded near-optimal target. By Lemma 4.11 there exists a bounded strategy $\delta \in \mathcal{H}^{\mathrm{u}}$, say $|\delta_k| \leq j$ a.s. for all k, such that

$$\rho_\ell\big(X + W(H \circ \delta)\big) \leq \pi(X) + \frac{\varepsilon}{2}.$$

Write $\eta := H \circ \delta$ and $Y := X + W(\eta)$.

Step 2: representing maps and integrability. Each δ_k is $\mathcal{F}_k$-measurable with $\mathcal{F}_k = \sigma(I_0, \ldots, I_k)$, so there exist Borel maps $f_k : \mathbb{R}^{r(k+1)} \to \mathbb{R}^d$ with $\delta_k = f_k(I_0, \ldots, I_k)$. Truncating each component of f_k to the interval $[-j, j]$ changes nothing $\mathbb{P}$-a.s. and makes f_k bounded everywhere. In particular every component of f_k lies in $L^1(\mu_k)$, where μ_k is the law of $(I_0, \ldots, I_k)$. This settles (F1).

Step 3: network approximation. By Theorem 4.1 applied componentwise, there exist networks $F_{k,m} \in \mathcal{NN}_{\infty,r(k+1),d}$ with $F_{k,m} \to f_k$ in $L^1(\mu_k)$ as $m \to \infty$. Set

$$\delta^m := \big(F_{k,m}(I_0, \ldots, I_k)\big)_{k=0,\ldots,n-1}.$$

Then $\delta_k^m \to \delta_k$ in $L^1(\mathbb{P})$ for every k. Passing to a subsequence (not relabelled) we may assume

$$\delta_k^m \longrightarrow \delta_k \ \mathbb{P}\text{-a.s.} \qquad \text{for all } k = 0, \dots, n-1$$

simultaneously. Since each $F_{k,m}$ belongs to $\mathcal{NN}_{\infty,r(k+1),d}$ and these families are nested with $\bigcup_M \mathcal{NN}_{M,r(k+1),d} = \mathcal{NN}_{\infty,r(k+1),d}$, there exist $M_m \uparrow \infty$ such that $\delta^m \in \mathcal{H}_{M_m}$.

Step 4: constrained strategies and objectives. Set $\eta^m := H \circ \delta^m$. Continuity of each $H_k(\omega, \cdot)$ and induction over k give $\eta_k^m \to \eta_k$ $\mathbb{P}$-a.s. for every k. Gains converge: $(\eta^m \cdot S)_T \to (\eta \cdot S)_T$ a.s. Upper semicontinuity of the $c_k(\omega, \cdot)$ yields

$$\limsup_{m\to\infty} C_T(\eta^m) \le C_T(\eta) \ \mathbb{P}\text{-a.s.},$$

and therefore, writing $Y_m := X + W(\eta^m)$,

$$\liminf_{m\to\infty} Y_m \ge Y \ \mathbb{P}\text{-a.s.}.$$

Furthermore, by (A6) and (A1), $Y_m \ge -\|Z\|_\infty$ a.s. for every m.

Step 5: conclusion. All conditions of Lemma 4.10 are now satisfied with $B = \|Z\|_\infty$, and it gives $\limsup_m \rho_\ell(Y_m) \le \rho_\ell(Y) \le \pi(X) + \varepsilon/2$. Finally, choose m_0 with $\rho_\ell(Y_{m_0}) \le \pi(X) + \varepsilon$. Since $\delta^{m_0} \in \mathcal{H}_{M_{m_0}}$, one has $\pi_M(X) \le \pi(X) + \varepsilon$ for all $M \ge M_{m_0}$.

Case $\pi(X) = -\infty$. Let $L \in \mathbb{R}$ be arbitrary. Choose $\delta \in \mathcal{H}^{\mathrm{u}}$ with $\rho_\ell(X + W(H \circ \delta)) \le L - 1$. Then, steps 1 to 5 above go through with the threshold L in place of $\pi(X) + \varepsilon$, yielding M with $\pi_M(X) \le L$. By choosing L lower and lower, and being $(\pi_M(X))_{M\in\mathbb{N}}$ nonincreasing, $\pi_M(X) \to -\infty = \pi(X)$. □

Remark 4.13. The proof uses (A1) and (A6), the standing continuity structure of the maps H_k and cost functions c_k, the OCE representation of ρ and Theorem 4.1. It does not use (A2) to (A5) (convexity of $\mathcal{H}$, the half-bounded Fatou property, L^0-boundedness of the gain set and closedness of the dominated gain cone). Those hypotheses remain needed for attainment in Chapter 3; they are not needed for value convergence. The full strength of (A1) is also not needed: the same proof works if Z is merely bounded from below with $\mathbb{E}[\ell(Z + c)] < \infty$ for every $c \ge 0$, the deterministic envelope in Lemma 4.10 being replaced by a random ℓ-integrable one. We do not pursue this weakening.

Combining Theorem 4.12 with the attainment theorem of Chapter 3 gives the punchline of this thesis.

Corollary 4.14. *Assume in addition* (A2) *to* (A5)*, so that the attainment theorem of Chapter 3 applies. Then there exists* $\delta^* \in \mathcal{H}$ *with* $\rho_\ell(-Z +$

$W(\delta^*)) = \pi(-Z)$, and for every $\varepsilon > 0$ there exist $M \in \mathbb{N}$ and $\delta^\varepsilon \in \mathcal{H}_M$ with

$$\rho_\ell\big(-Z + W(H \circ \delta^\varepsilon)\big) \le \rho_\ell\big(-Z + W(\delta^*)\big) + \varepsilon.$$

Proof. The existence of δ^* follows from the attainment theorem of Chapter 3, and the ε-statement is Theorem 4.12. □

In one sentence: under (A1) to (A6), neural network strategies are ε-optimal for an *attained* optimum.

4.4 From theory to algorithm

Theorem 4.12 justifies replacing the strategy class $\mathcal{H}^{\mathrm{u}}$ by the network classes $\mathcal{H}_M$; this section records why that replacement makes the hedging problem computable. For an OCE one has $\rho_\ell(Y) = \inf_{w\in\mathbb{R}}\{w + \mathbb{E}[\ell(-Y - w)]\}$, so with $Y = -Z + W(H\circ\delta^\theta)$, where $\delta^\theta \in \mathcal{H}_M$ is determined by the finitely many network weights θ of the maps F_{θ_k}, the approximate problem (2) becomes

$$\pi_M(-Z) = \inf_{(w,\theta)} \Big\{ w + \mathbb{E}\big[\ell\big(Z - W(H \circ \delta^\theta) - w\big)\big] \Big\}.$$

This is a finite-dimensional minimization over the pair (w, θ), jointly and without any inner optimization. In practice the expectation is estimated by minibatches of simulated (or historically calibrated) sample paths of the information process I, the gradient in (w, θ) is computed by backpropagation through the semi-recurrent parametrization $\delta_k = F_{\theta_k}(I_0, \dots, I_k, \delta_{k-1})$, and the minimization is carried out by stochastic gradient descent. For the entropic risk measure the inner minimizer is explicit,

$$w^* = \frac{1}{\lambda} \log \mathbb{E}\big[e^{\lambda(Z - W(H\circ\delta^\theta))}\big],$$

so one minimizes over θ alone; cf. Bühler et al. [1, Section 3.2].

This thesis contains no numerical experiments of its own. Bühler et al. [1, Section 5] train these networks and backtest the resulting hedges: in their experiments the learned strategies achieve a substantial reduction of the hedging risk relative to the unhedged position and remain competitive with classical model-based benchmarks in the presence of transaction costs, where the classical strategies are no longer optimal. One structural caveat must be stated explicitly: the networks are trained exclusively on historic data or on simulated data calibrated to historic observations. A backtest evaluates the hedge on the same statistical regime it was trained on, and past performance is not indicative of future results; this limitation is inherent to the method and not an implementation detail. Finally, for general convex risk measures beyond OCEs, the robust representation leads to a minimax problem over two networks (Bühler et al. [1, Section 4.5]); this extension is not pursued in this thesis.

4.5 Strategies need not converge

Theorem 4.12 and Corollary 4.14 are statements about *values*: the network values converge to $\pi(-Z)$, and near-minimizers are ε-optimal. We close the chapter by observing that nothing of this kind holds for the strategies themselves.

Example 4.15. Let $S \equiv s_0$ be constant, $C_T \equiv 0$ and $\mathcal{H} = \mathcal{H}^{\mathrm{u}}$ (no constraints). Then $W(\delta) = 0$ for every δ, so every strategy is optimal for every Z and $\pi(-Z) = \rho_\ell(-Z)$. Network near-minimizers may oscillate indefinitely between distinct elements of $\mathcal{H}_M$ without converging in any topology. The same phenomenon occurs with two identical assets and zero costs: the strategies $(\delta, 0)$ and $(0, \delta)$ produce the same gain, and minimizers may alternate between them.

Remark 4.16. Theorem 4.12 and Corollary 4.14 should therefore be read as value statements only. Without strict convexity of the loss function ℓ and a nonredundancy condition on S, optimal strategies need not be unique, and a sequence of network near-minimizers $\delta^M \in \mathcal{H}_M$ need not converge to an optimal strategy δ^* in any reasonable topology. What the theory guarantees is that the *risk* of δ^M approaches the optimal risk; which minimizer the training procedure selects is not determined by the value convergence.

Bibliography

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[2] K. Hornik, Approximation capabilities of multilayer feedforward networks, *Neural Networks* 4(2), 251-257, 1991.